

External Internship

Thermal bonding of PMMA substrates

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1. Introduction

The topic of this project is "Developing a bonding chamber suitable for thermal bonding of PMMA substrates for the purpose of creating microfluidic devices" and the project was conducted at the physics department of the National University of Singapore (NUS). These microfluidic devices are channels with a width at the (sub-)micrometer scale encased in a transparent polymer. Liquids can flow through these channels, and the transparency of the polymer allows for analysis of the liquid. Various types of research are currently being done with such microfluidic devices, which makes this a relevant project.

Several methods can be used to manufacture microfluidic devices. One such method, which is inexpensive with a high throughput, is nanoimprint lithography (NIL).^[1] The process is depicted in Figure 1. In this process a stamp with a structured surface is used, upon which a polymer substrate is placed (Step 1). By heating to a temperature above the glass transition temperature T_g of the polymer and applying a pressure p , the polymer surface conforms to that of the stamp (Step 2). Subsequently, the temperature is lowered while maintaining pressure (Step 3), after which the polymer with the imprinted structure can be removed (Step 4). Structures with dimensions smaller than a micron can be imprinted using this method. In this project a nickel stamp is used; the polymer used is Poly(methyl methacrylate) (PMMA), which has a T_g of 105°C .

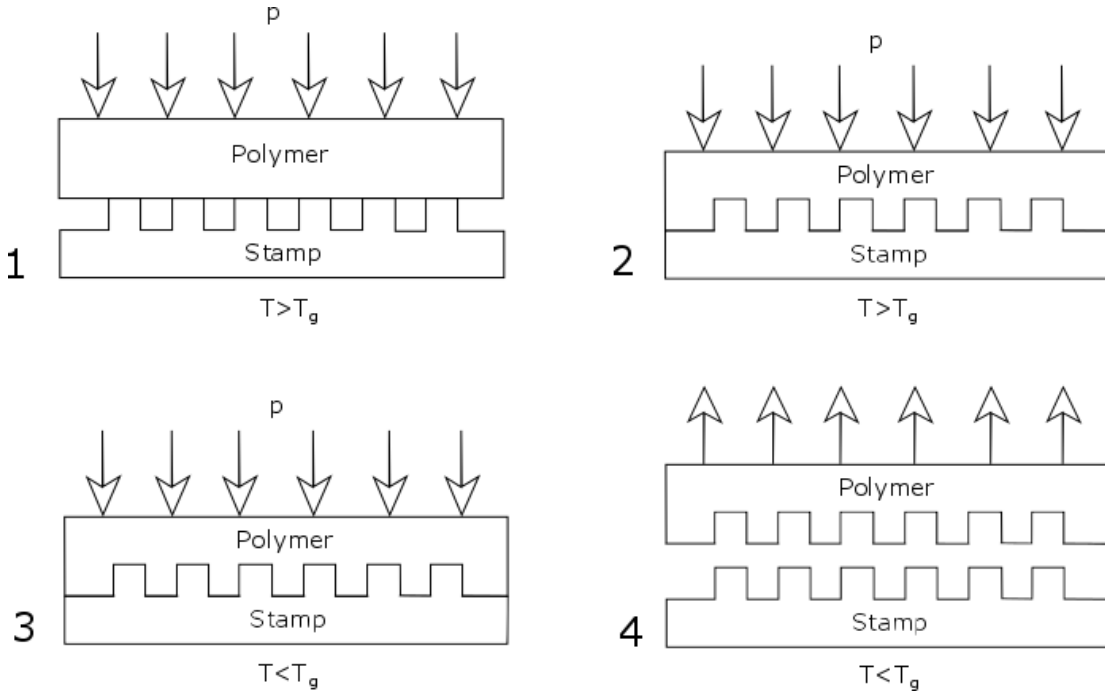


Figure 1: The process of nanoimprint lithography.

At the CIBA-lab (Centre for Ion Beam Applications) of the National University of Singapore, a nanoimprinter is available. It can be seen in Figure 2a. The stamp and the PMMA substrate are placed on the sample stage, after which the stage is covered by a Kapton film, as seen in Figure 2b. The sample stage can then be set into position. Upon starting the the imprinter, the stage will be pressed against a rubber seal. The Kapton film then serves as a means to equally distribute the

pressure over the PMMA substrate. This has been illustrated in Figure 3. After the previously described imprinting process is complete, the imprinted sample can be removed from the machine.

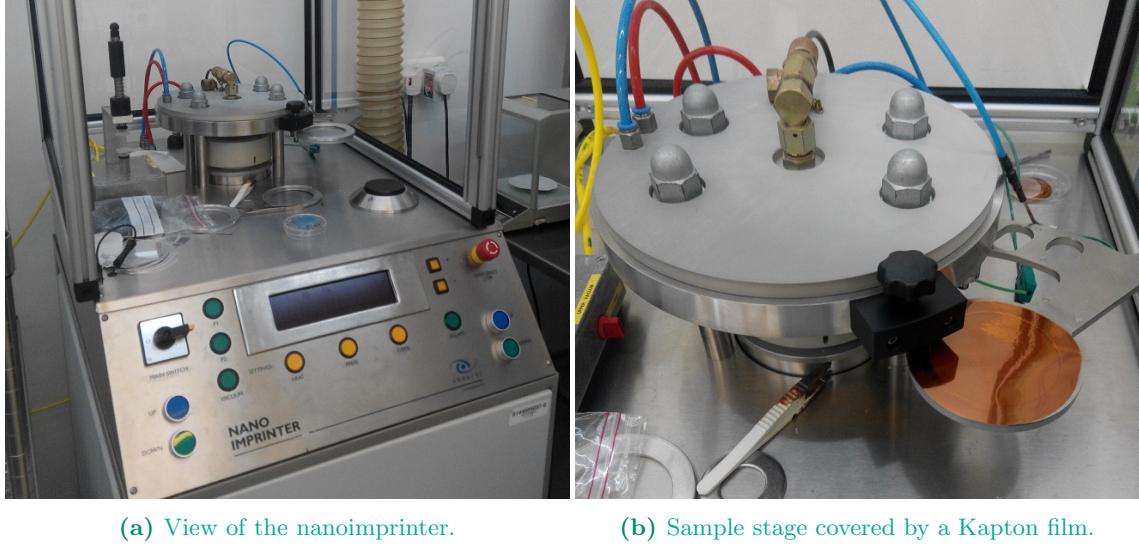


Figure 2: Commercial nanoimprinter available in the clean room of the CIBA-lab.

After the open micro-channels have been formed in the PMMA substrate, the substrate has to be bonded to a second substrate to close the micro-channels. This thermal bonding process is very similar to the imprinting process shown in Figure 1. The difference is that this process uses two PMMA substrates instead of a stamp and a PMMA substrate, as seen in Figure 3. The temperature and pressure are increased and the substrates are given time to bond. After cooling down, the pressure is released and the substrates are bonded. In order to ensure successful bonding, pressure and temperature conditions need to be high enough. However, conditions which are too high will cause the micro-channels to collapse. Previous studies have shown that at the T_g of PMMA (105°C), a pressure of 1.2 bar can be used to bond two substrates without deformation or channel collapse.[2]

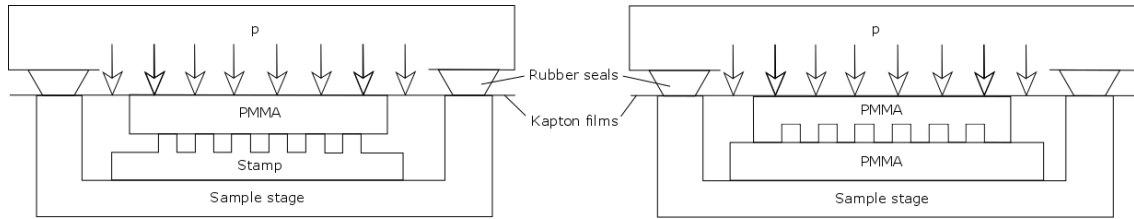


Figure 3: Schematic drawing of the nanoimprinter, as used for imprinting (left) and bonding (right).

While the nanoimprinter could perform this bonding process, it is unable to accurately regulate pressure in the range of 10 bar and below. It is therefore unsuitable for thermal bonding of substrates with micro-channels. For this project, it has been proposed to design and develop a new device with which temperature and pressure can be regulated at values more appropriate to the bonding process.

In this report, the nanoimprinting analysis will be discussed first. After that, the design process of the bonding chamber will be explained. Bonding result obtained using the bonding chamber will be discussed, and finally the conclusions will be summarized and recommendations will be given.

2. Nanoimprint analysis

The nanoimprinter has been used by several people. However, the input values that have been used vary greatly. Therefore, several imprints have been made using a nickel stamp to determine the ideal environmental conditions for imprinting. The results were observed with an optical microscope and some examples can be seen in Figure 4. The micro-channels seen here are $20\mu\text{m}$ wide and were created by imprinting for 30 seconds.

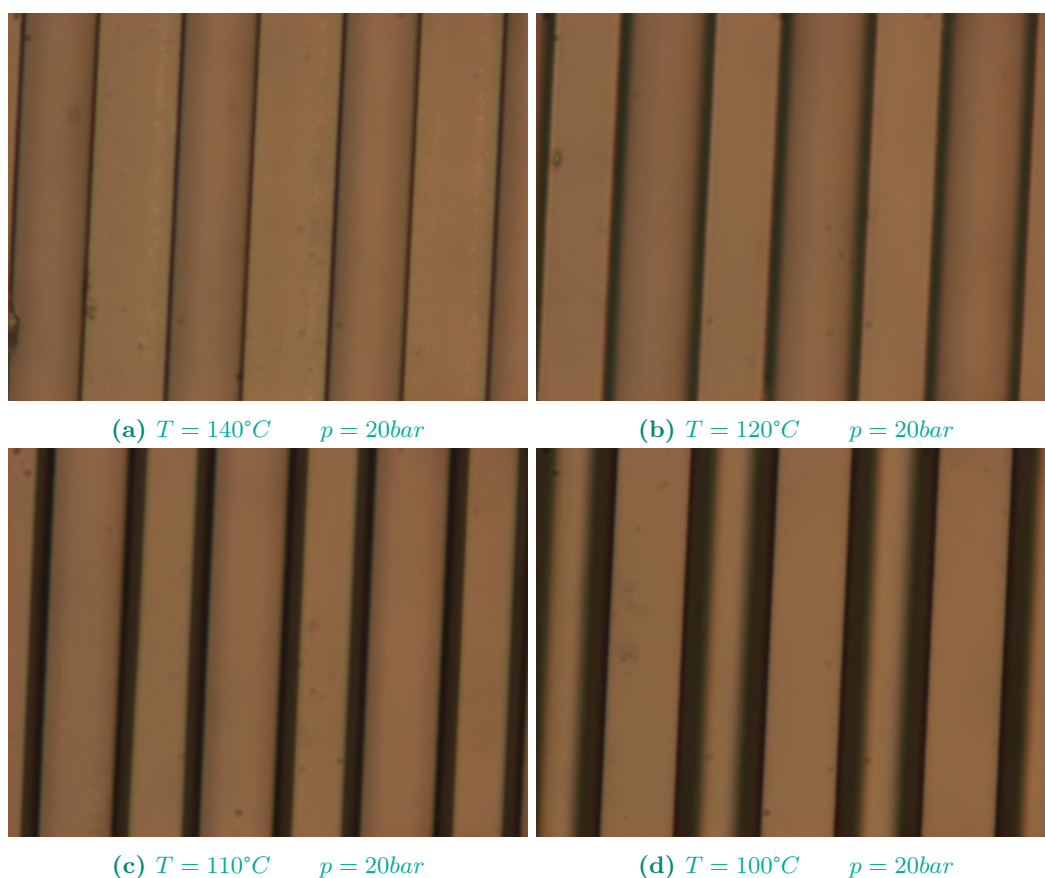


Figure 4: Micro-channels made at different temperatures. Images made with an optical microscope with a magnification factor of 50.

A schematic representation of the above results is shown in Figure 5. It can be seen that for lower temperatures the imprinting is not fully completed. The imprint in Figure 4a does have straight walls, like the left example in Figure 5.

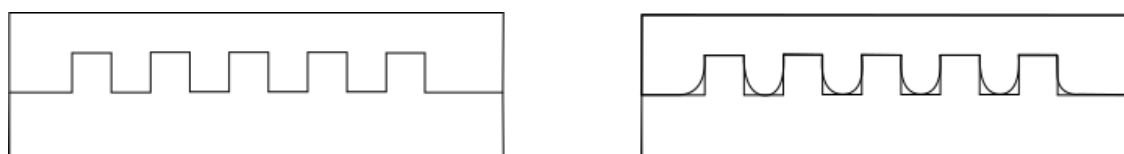


Figure 5: Schematic representation of perfect (left) and incomplete (right) imprinting results.

After conducting further tests on other pressure levels, acceptable pressure-temperature configurations have been determined. For pressure increments of 10 bar, these configurations are shown below:

$$p = 10 \text{ bar} \quad T = 140^\circ\text{C}$$

$$p = 20 \text{ bar} \quad T = 130^\circ\text{C}$$

$$p = 30 \text{ bar} \quad T = 120^\circ\text{C}$$

It is possible to deviate from these input values. Using higher values will produce practically the same results, but might not be as time-efficient. When using values lower than the ones shown above, one should be aware that the results may not be completely straight-walled.

During imprinting, the temperature is first increased to the setpoint, followed by the pressure¹. After this, pressure and temperature are maintained for a set amount of time; a duration of 30 seconds has been used during this project. Then temperature is brought down well below T_g and finally pressure is released.

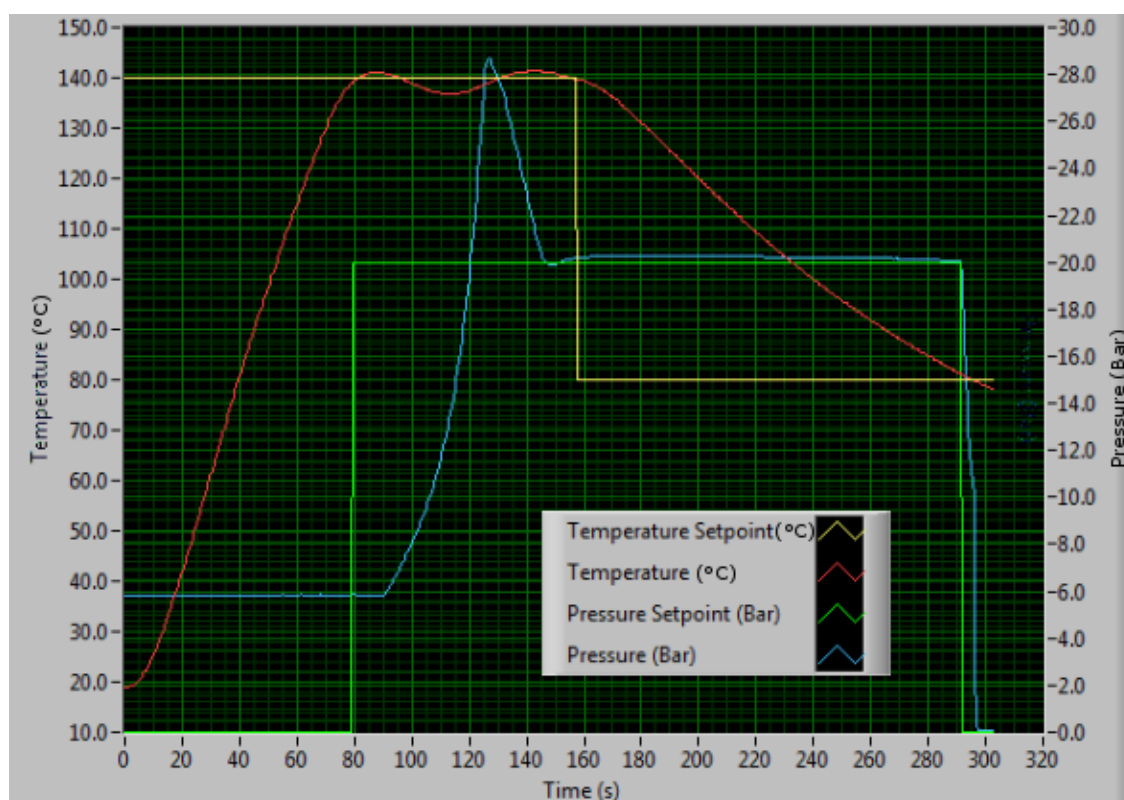


Figure 6: Pressure and temperature levels and setpoints during the imprinting process.

As seen in Figure 6, the cool-down phase is the most time-consuming. In order to minimize the total time needed for imprinting, it is essential to keep temperatures low. Therefore, using a pressure of 10 bar with a temperature of 140°C will take the longest of the recommended settings, but this is only a concern when time is of the utmost essence.

¹The pressure always overshoots the setpoint considerably.

3. Chamber design

A different form of bonding uses UV-radiation to bond two substrates together. Although the scope of this project only included thermal bonding, it was important for the bonding chamber to have the option available for possible future application. In order to use UV-radiation during the bonding process, the top of the bonding chamber needs to be transparent. Additionally, this will simplify proper alignment of the two substrates. For this purpose, a flange with a built-in window was available. It was supposedly able to withstand pressures in the range of interest and the dimensions allowed for sufficiently large samples to be bonded, therefore the rest of the chamber was designed with this top component in mind. The component can be seen in Figure 7.

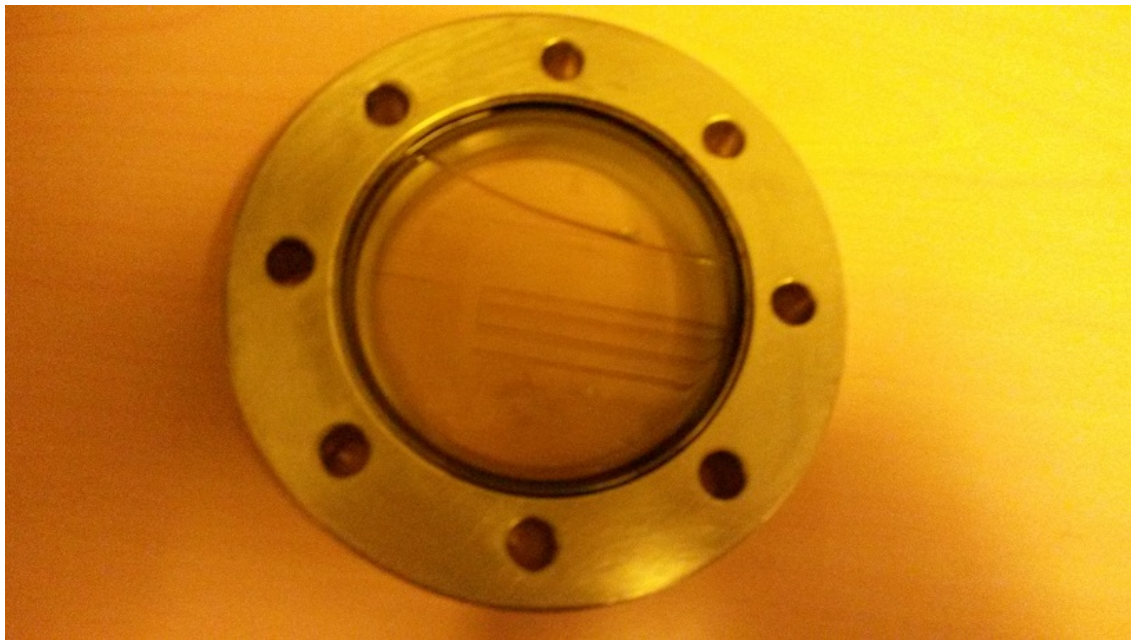


Figure 7: Top component of the bonding chamber.

Due to a minor defect in the window, fracture occurred during initial testing. The component was then replaced by a solid stainless steel flange with the same dimensions, since UV-radiation was not necessary at this point in time.

With the dimensions of the window flange known, a rough design was made, which is shown in Figure 8 to illustrate the working principle of the bonding chamber. The chamber consists of three components, clamped together with bolts and separated by rubber seals. Between the bottom and middle component, a Kapton sheet is held in place by two flat rubber rings. The space above the sheet is pressurized through an opening in the middle component. The space below the sheet serves as the sample stage; if necessary a vacuum pump can be connected through an opening in the bottom component. Heating of the sample is done by placing the chamber on a hot plate. Since there is a temperature difference between the hot plate and the sample stage, calibration is necessary to estimate the working temperature. To heat the sample at 105°C , the hot plate has to be set to 115°C .

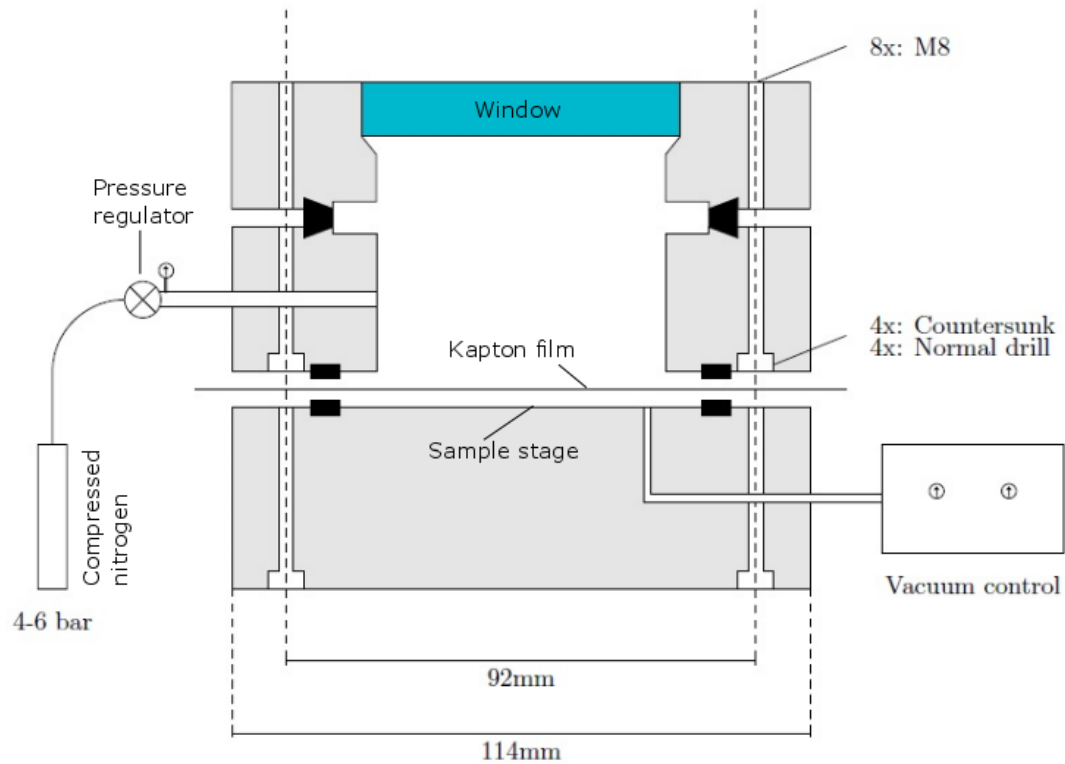


Figure 8: Initial design of the bonding chamber, dimensions were not yet fully detailed.

The chamber has been modeled using NX7.5; the result can be seen in Figure 9. For detailed technical drawings of the two fabricated components, refer to the appendix.

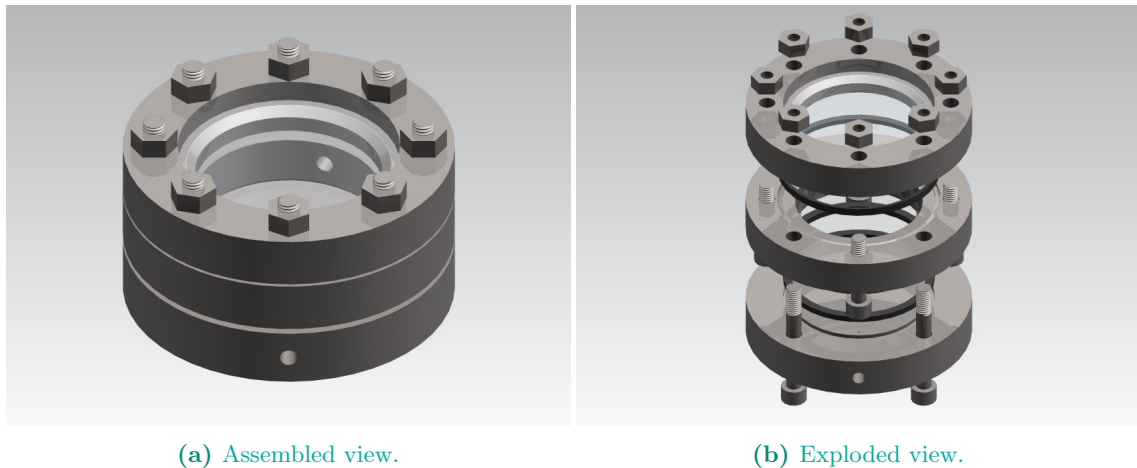


Figure 9: Bonding chamber modeled in NX7.5.

After consulting with the workshop, the decision fell on welding quarter inch tubes into the inlet and outlet of the bonding chamber, as a means of connecting it to the nitrogen line or the vacuum installation. The final product can be seen in Figure 10.

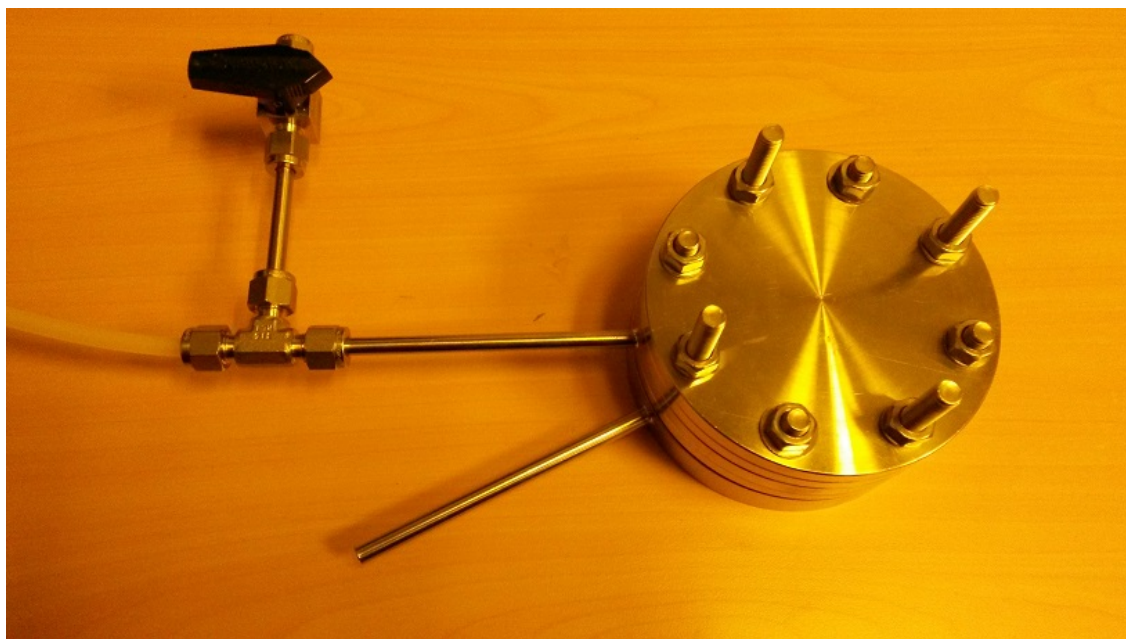


Figure 10: Finished bonding chamber, with solid stainless steel flange as top component.

Initially, circular grooves were present on the sample stage, which caused problems for the bonding process. A lapping machine was used to reduce the surface roughness of the sample stage, giving it a mirror finish. This can be seen in Figure 11.

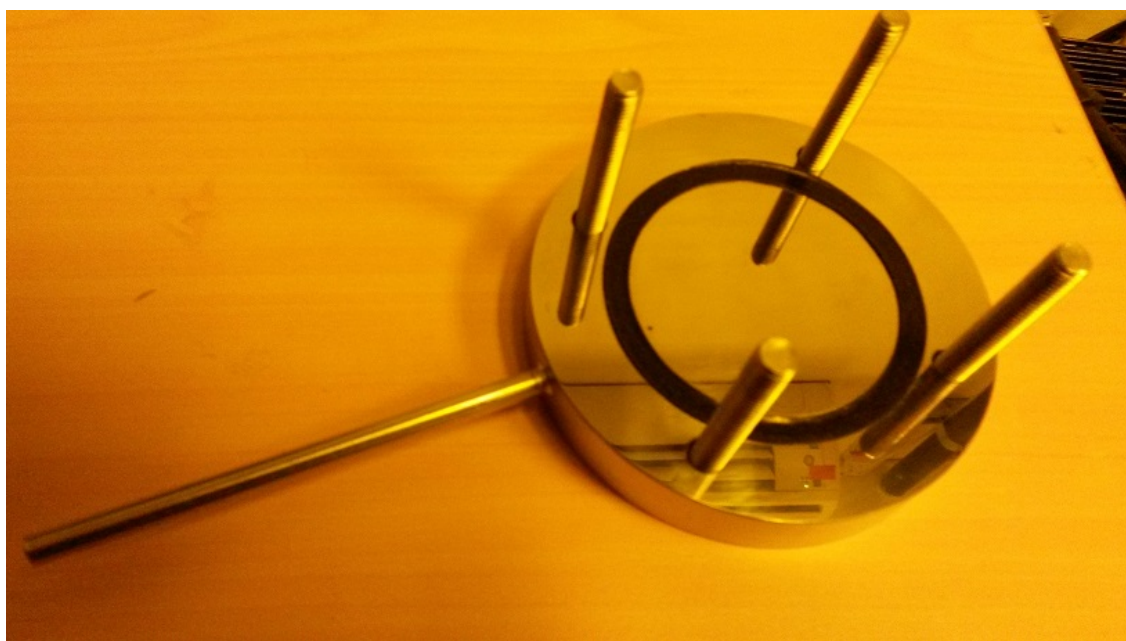


Figure 11: Bottom component, after lapping machine treatment.

The bonding chamber is connected by a flexible tube to a pressure regulator, with which the desired pressure can be set. Several valves have been set in place to ensure safe pressure levels, as well as a valve for venting the system. The regulator setup can be seen in Figure 12. The pressure gauges currently mounted to the regulator allow for pressures up to 2 bar.



Figure 12: Pressure regulator.

At the end of the project, another window flange became available and the solid stainless steel flange was replaced. Unlike the previous window, this one did not have any defects and was able to withstand the desired pressure. As shown in Figure 13, it is now possible to get a view of the inside of the bonding chamber, which is useful for checking for proper alignment before starting the bonding process.

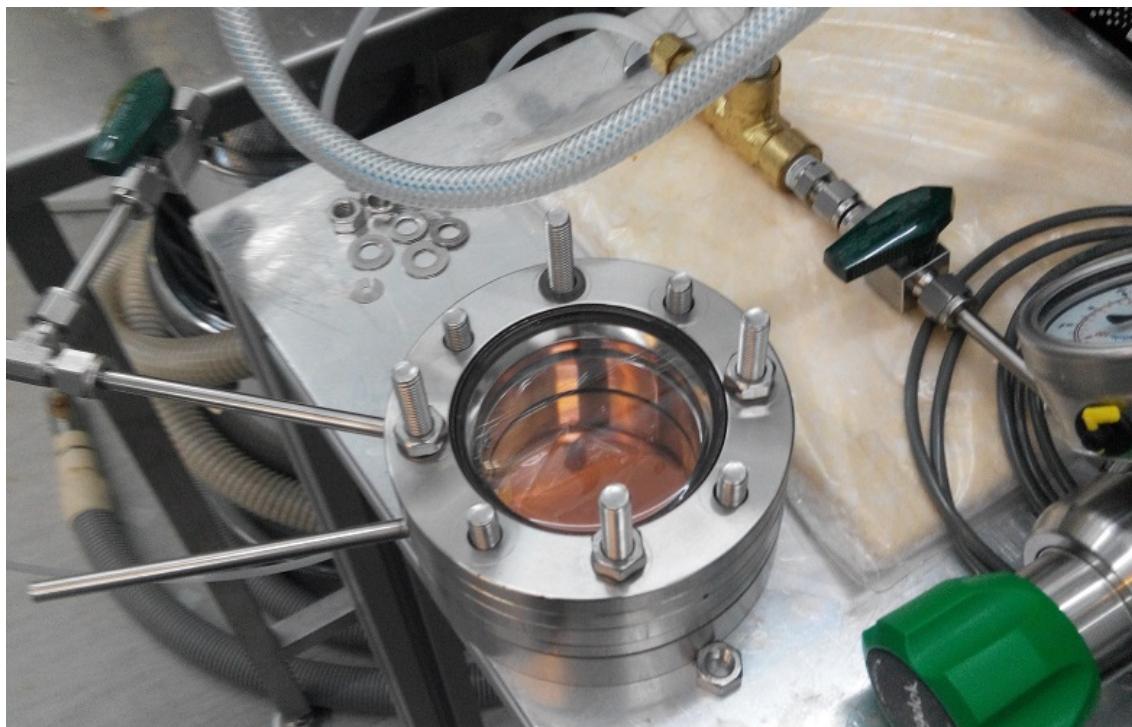


Figure 13: Finished bonding chamber, with window flange as top component.

4. Bonding experiments

Initial testing of the bonding chamber proved successful. Two imprinted PMMA samples were taken, both containing $20\mu\text{m}$ micro-channels, and were positioned at a 90 degree angle with respect to each other. The result can be seen in Figure 14b.

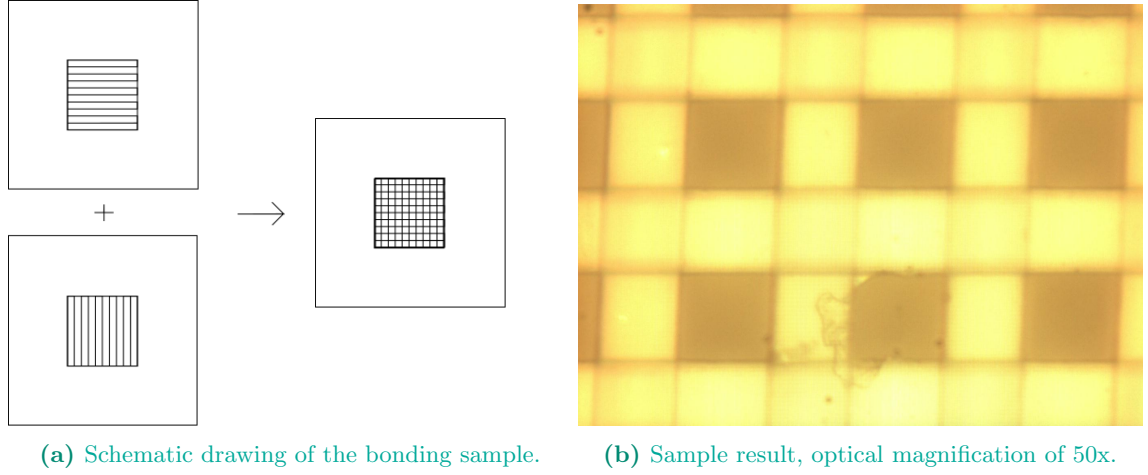


Figure 14: Initial bonding experiment.

The bonded sections can clearly be distinguished from the micro-channels by the darker hue. It can be seen that the micro-channels remain open and that the rest has been bonded, as is required for a functioning microfluidic device.

Upon further testing, the main problem is found to be the presence of air bubbles where bonding has not taken place. Major causes of this are dust particles and surface irregularities. The presence of dust particles was diminished by eventually moving the bonding process to the clean room.

To introduce a liquid into the micro-channels, an access hole needs to be present. Drilling was used to create holes in the PMMA substrates. However, the area around the drill holes was deformed to such an extent that successful bonding was prevented. An inclination is formed around the drill hole, as shown in Figure 15. An inclination with a height of $50\mu\text{m}$ can be seen in Figure 16a, the result this has for bonding can be seen in Figure 16b. To prevent extensive deformation around drill holes, a clamping device was created.



Figure 15: Schematic representation of the cross-section of a deformed substrate after drilling.

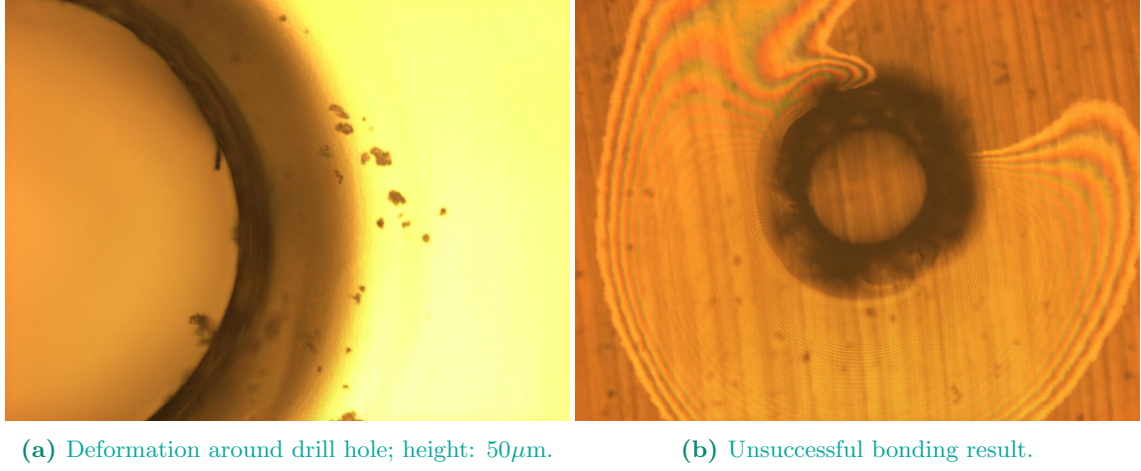


Figure 16: Results obtained after drilling holes without clamp.

4.1 Sample preparation

Using a laser cutter, the basis for a clamping device was cut from an acrylic sheet, consisting of two square pieces of acrylic with holes in each corner to fit magnets. In the middle, two half millimeter holes were cut to serve as guidance for the drill. The first version of the device can be seen in Figure 17a. Tape has been used to keep the magnets in place.

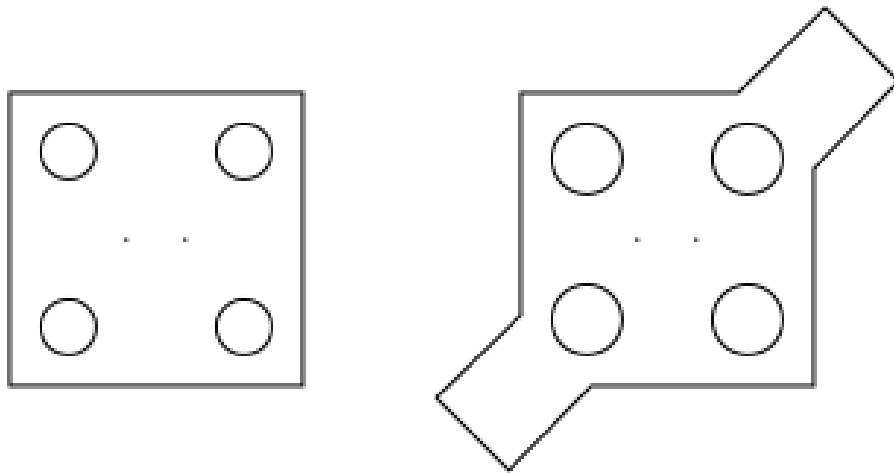
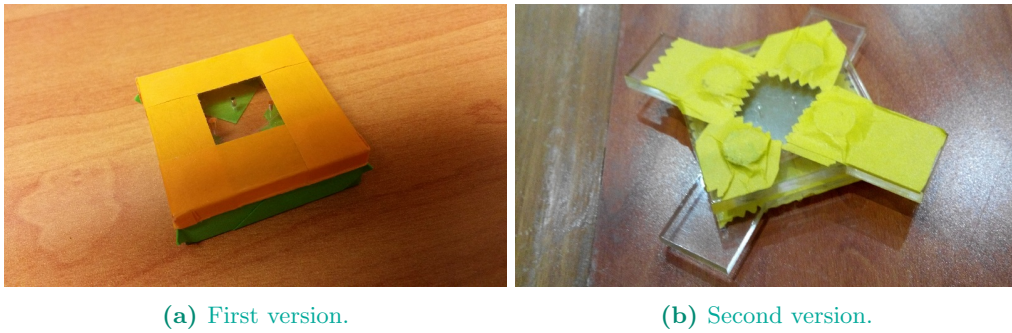


Figure 17: Magnetic clamping devices.

The concept is to clamp the PMMA samples between the two sheets when drilling, thereby reducing the deformation around the drill hole. Better results are obtained when using an additional acrylic sheet as a sacrificial layer. This setup is shown in Figure 18. Deformation around the drill holes was slightly decreased using the first clamping device. Since the magnets lacked the force required to effectively clamp the samples in place, a second clamping device was developed with stronger magnets. Additionally, the design was altered to simplify the separation of the two clamp pieces. This can be seen in Figure 17b.

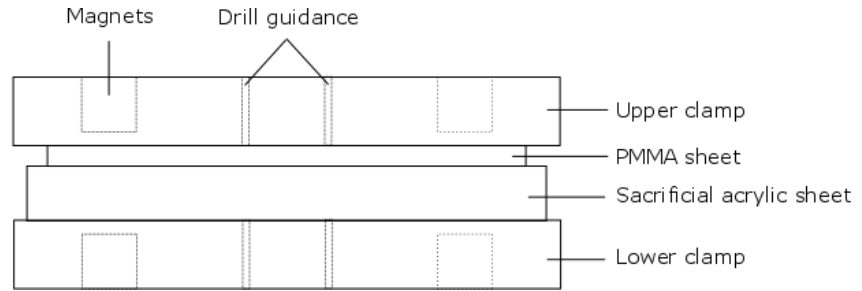
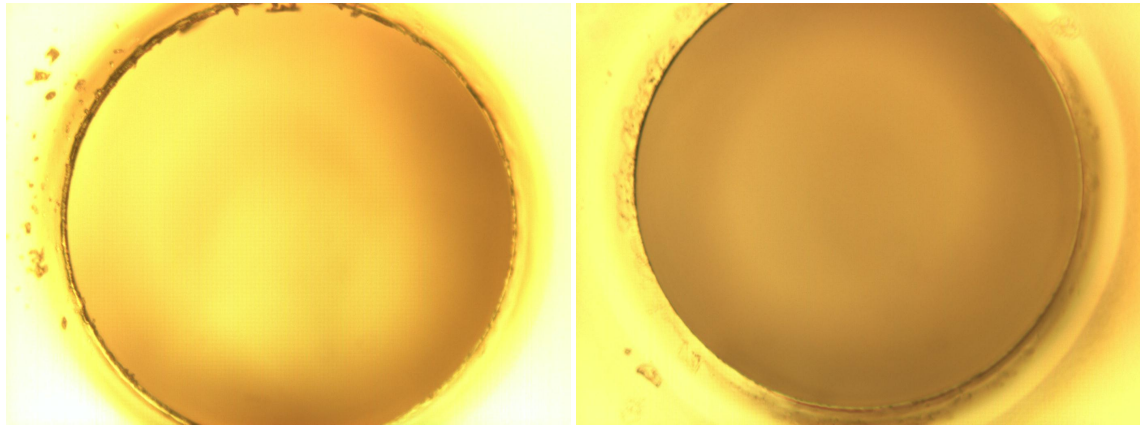


Figure 18: Schematic drawing of the proper usage of the clamping device.

The results obtained from using these clamps can be observed in Figure 19, the images here show the area around the drill hole at the side of the PMMA sheet opposite to where the drill entered. The stronger clamp proved to be the most effective, as shown in 19b. Deformations were usually in the range of $5\mu\text{m}$ or lower. The weaker clamp results, shown in Figure 19a, had deformations around $20\mu\text{m}$ to $30\mu\text{m}$ above the sheet surface.



(a) Weaker clamp result, height: $20\mu\text{m}$.

(b) Stronger clamp result, height: $5\mu\text{m}$.

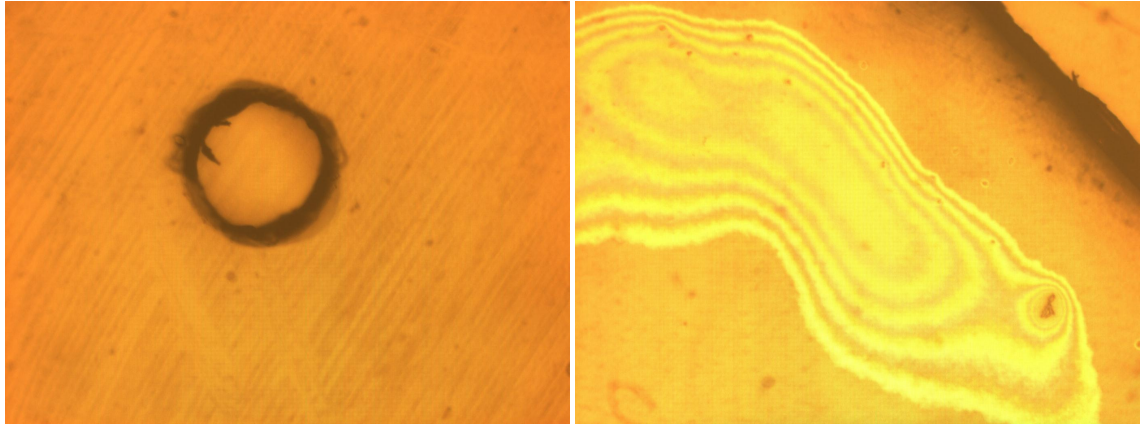
Figure 19: Drill holes obtained using clamping devices.

Another form of sample preparation that was considered was surface treatment with a plasma cleaner. This removes impurities from the surface and increases the hydrophilicity. However, no noticeable differences were observed. Effects on the bonding strength of the samples could be interesting for future investigation.

4.2 Bonding results

For the majority of the bonding experiments, pressure and temperature values mentioned in literature were used. A temperature of $T = 105^{\circ}\text{C}$ and a pressure of 1.2 bar. At higher temperature values, deformation is observed. No visible differences have been observed from using higher pressures, e.g. 1.5 bar. The PMMA sheet used during this project had a thickness of $375\mu\text{m}$.

In the Figure 20, a PMMA substrate with a single hole drilled into the middle was bonded to another substrate. Bubble formation around drill holes has been greatly reduced, as seen when comparing Figure 20a to Figure 16b. However, bubbles are occasionally still formed near the edges of the bonded chip. An example of this is shown in Figure 20b. Since this is generally not the working area of the chip, it is not a big issue.



(a) Bonded sample without bubble near drill hole. (b) A bubble near the edge of the bonded chip.

Figure 20: Various bonding results.

To test whether the produced chips work properly, experiments have been conducted on two chips with different channel widths. One chip with channel width of $20\mu\text{m}$ and one with a width of 200nm . A droplet of isopropanol (IPA) was placed on the access holes, like the ones seen in Figure 21; through capillary forces, the IPA was led through the micro/nano-channels. The tests were successful, as can be seen in Figures 22 and 23 where the fluid propagation has been visualized.



Figure 21: Pre-bonding substrate alignment.

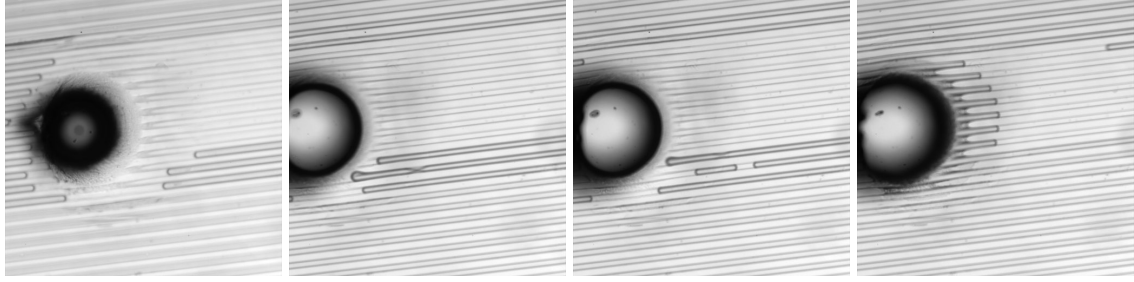


Figure 22: IPA propagating/evaporating in micro-channels.

In these pictures, the channels with darker colors are filled with air. Channels with lighter colors are filled with IPA. Both cases showed similar results. After introducing the IPA, the liquid quickly filled the channels. Shortly after, the IPA evaporated and the channels were seen slowly filling up with air. Refer to the appendix for additional images of this process. Propagation through the micro-channels from Figure 22 happened much faster, therefore no additional images of this experiment are provided.

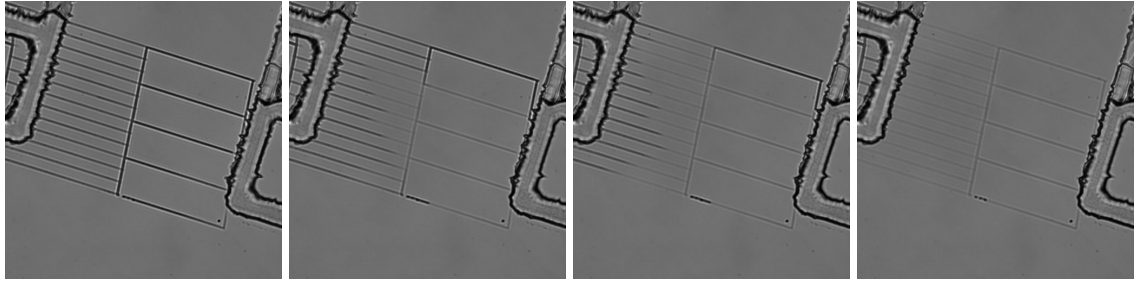


Figure 23: IPA propagating through nano-channels.

5. Discussion

5.1 Conclusions

During this project, a bonding chamber has successfully been designed and developed. Tests have proven it to be a working product. A way to ensure proper bonding around drill holes by reducing the surface irregularities has been investigated, with positive results. Finally, the functionality of the chip after bonding has been tested by observing the flow of IPA through the micro/nano-channels. Footage showed that the chips were properly bonded and functional. Further studies will need to be done to optimize the bonding process, a few recommendations are presented below.

5.2 Recommendations

The experiments in this project only focused on whether proper bonding occurred. To optimize the process, bonding strength will need to be evaluated for various temperature, pressure and exposure times. One possible way to do this is to pump a liquid into the channels and verify which sample can withstand the highest pressure without substrate separation. It may also be worth investigating whether plasma cleaning the substrates before bonding makes a difference in terms of bonding strength.

Currently the temperature of the sample is estimated by measuring the temperature over time before conducting the bonding. To circumvent extensive calibrations, a thermocouple can be built into the bonding chamber. This will give a more accurate representation of the temperature as a function of time. To reduce the time it takes to heat up the chamber to the appropriate temperature, it could be worth minimizing the thickness of the bottom component in the current or potential future designs.

The use of vacuum has yet to be investigated. It could be interesting to test whether better bonding occurs under vacuum conditions.

6. Appendix

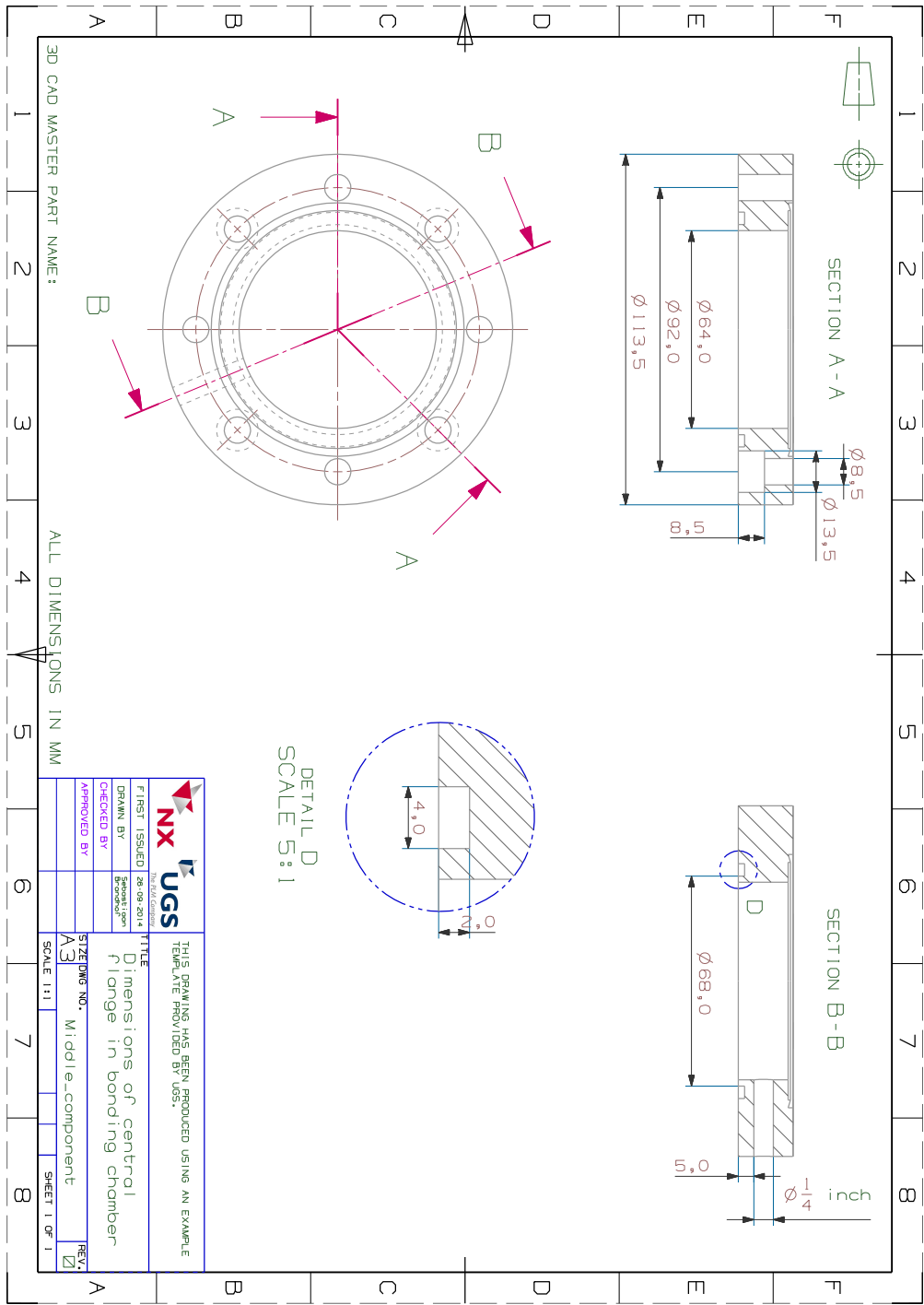


Figure 24: Technical drawing for the middle component of the bonding chamber.

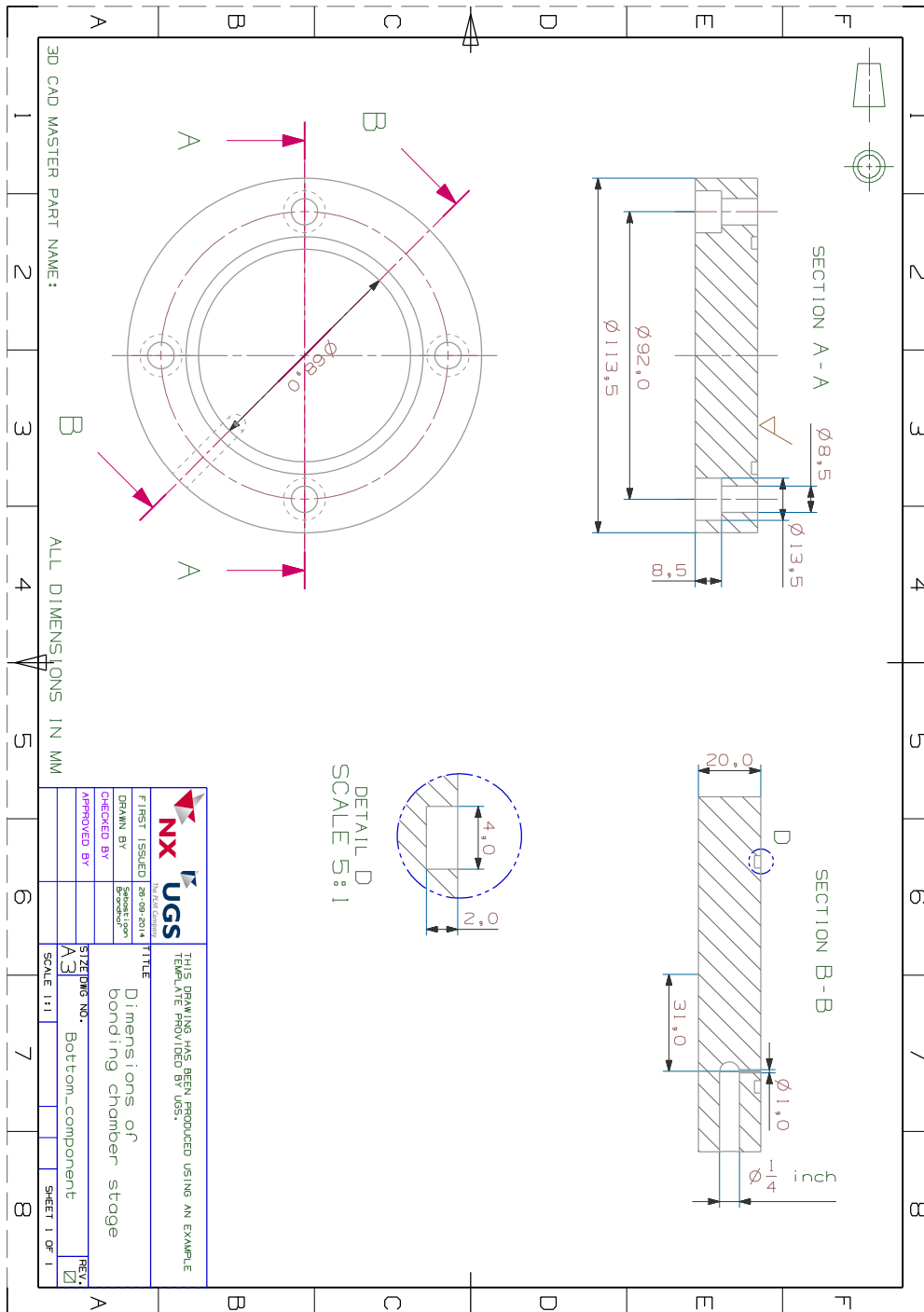


Figure 25: Technical drawing for the bottom component of the bonding chamber.

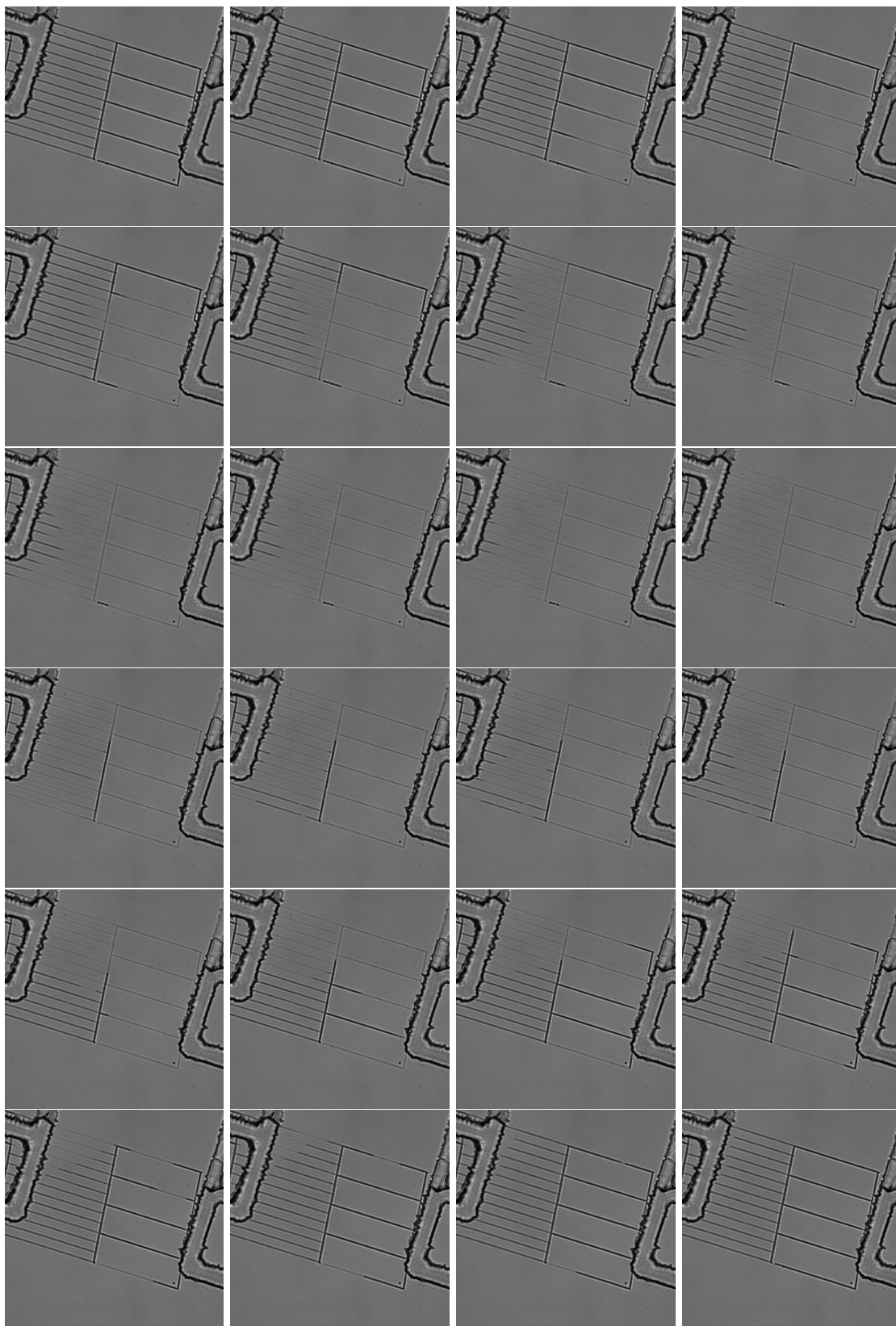


Figure 26: IPA propagating through nano-channels, subsequently evaporating.

7. Bibliography

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- [2] P.G. Shao, J.A. van Kan, L.P. Wang, K. Ansari, A.A. Bettiol, F. Watt, *Rapid prototyping of micro/nano poly (methyl methacrylate) fluidic systems using proton beam writing*. Nuclear Instruments and Methods in Physics Research B 260 (2007) 362-365.